Performance Analysis of Software Defined Networks (SDN) Through Simulation Using Mininet and OpenFlow

A CAPSTONE PROJECT REPORT

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IN

ECE

Submitted by

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DECLARATION

WE PALAJI OBULESH(192312243),A VENKAT DHARAHAS(192412570) of the **ECE,** Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **'Performance Analysis of Software Defined Networks (SDN) Through Simulation Using Mininet and OpenFlow'**is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai

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BONAFIDE CERTIFICATE

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Contribution (%)
Student 1 – G. Devendranath Reddy (192425204)	Project planning and SDN system design	Designed the SDN architecture and network topology using Mininet	Tested basic network connectivity and topology performance	Introduction, Problem Analysis, Solution Design	**35%**
Student 2 – PALAJI OBULESH (192312243)	Mininet simulation and OpenFlow implementation	Developed the Mininet topology and configured OpenFlow switches	Tested throughput, delay, and packet transmission	Implementation and Results	**35%**
Student 3 – I. A VENKAT DHARAHAS (192412570)	System integration and performance analysis	Integrated the SDN controller, switches, and monitoring components	Tested complete network performance and analyzed results	Conclusion, Learning Outcomes, References	**30%**
Total	—	—	—	—	**100%**

ABSTRACT

Software Defined Networking (SDN) is a modern networking approach that separates the control plane from the data plane. This separation allows network administrators to manage and control the network through centralized software-based controllers. SDN provides flexibility, programmability, simplified network management, and efficient resource utilization.

This project focuses on analyzing the performance of Software Defined Networks through simulation using Mininet and OpenFlow. Mininet is used to create a virtual network environment consisting of hosts, switches, and links, while OpenFlow is used as the communication protocol between the SDN controller and network switches.

Different network configurations are simulated by changing the number of hosts and switches. Performance parameters such as throughput, delay, packet loss, and network utilization are analyzed. The simulation helps in understanding how network performance changes when the size and traffic load of the network increase.

The results show that SDN provides efficient centralized control and flexible network management. However, network performance may decrease when the controller handles a large number of switches and flow requests. The project demonstrates the importance of proper controller placement, network configuration, and resource management.

Keywords: Software Defined Networking, SDN, Mininet, OpenFlow, Network Performance, Throughput, Delay, Packet Loss.

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CHAPTER 1: INTRODUCTION

1.1 Introduction to Software Defined Networking

Computer networks are an important part of modern communication systems. Traditional networks use network devices such as routers and switches, where both the control functions and data forwarding functions are integrated within the devices.

Software Defined Networking (SDN) introduces a new approach to network management. SDN separates the network into two main components:

- **Control Plane** – Responsible for making network decisions.
- **Data Plane** – Responsible for forwarding data packets.

The SDN controller manages the network centrally and communicates with network devices using protocols such as OpenFlow.

1.2 Need for the Study

Traditional networks can be difficult to manage when the network becomes large and complex. Configuration changes often need to be performed individually on different network devices.

SDN simplifies this process by providing centralized control. However, the performance of an SDN network depends on several factors, including:

- Number of hosts
- Number of switches
- Network traffic
- Controller performance
- Flow setup time

Therefore, performance analysis is necessary to understand the behavior of SDN networks under different configurations.

1.3 Objectives

The main objectives of this project are:

1. To understand the architecture of Software Defined Networking.

2. To simulate an SDN environment using Mininet.
3. To use OpenFlow communication between the controller and switches.
4. To analyze network performance under different configurations.
5. To measure throughput, delay, and packet loss.
6. To compare the performance of different network sizes.
7. To provide recommendations for improving SDN performance.

1.4 Scope of the Project

The project focuses on the simulation of SDN networks in a virtual environment. Mininet is used to create hosts and OpenFlow-enabled switches.

The project does not require physical networking hardware. Different network topologies can be created and tested using software simulation.

CHAPTER 2: PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Problem Identification

Traditional networks have several limitations, including complex management and limited flexibility. As the number of devices increases, managing individual network devices becomes more difficult.

SDN solves many of these problems by providing centralized control. However, the centralized controller can become a performance bottleneck when the network grows.

The main problem identified in this project is:

How does the performance of an SDN network change when the number of hosts, switches, and network traffic increases?

2.2 Existing System

In traditional networks:

- Control and data forwarding functions are combined.
- Configuration is performed on individual devices.
- Network management becomes difficult for large networks.
- Dynamic changes require manual configuration.
- Centralized monitoring is limited.

2.3 Limitations of the Existing System

The major limitations include:

- Complex configuration
- Limited flexibility
- Difficult network management
- Slow adaptation to network changes
- Higher administrative effort

2.4 Proposed System

The proposed system uses Software Defined Networking with Mininet and OpenFlow.

The system includes:

- Multiple virtual hosts
- OpenFlow-enabled switches
- SDN controller
- Different network topologies
- Performance monitoring tools

The controller manages the switches centrally and determines how network traffic should be forwarded.

2.5 Performance Parameters

The following parameters are used for analysis:

Throughput

Throughput represents the amount of data successfully transmitted through the network in a given time.

Delay

Delay represents the time required for data packets to travel from the source to the destination.

Packet Loss

Packet loss occurs when packets fail to reach their destination.

Network Utilization

Network utilization indicates how efficiently the available network resources are used.

2.6 Problem Analysis

When the number of devices increases:

- More flow requests are generated.
- The controller processes more information.
- Flow setup time may increase.

- Network delay may increase.
- Overall performance may decrease.

Therefore, the performance of SDN must be analyzed under different network conditions.

CHAPTER 3: SOLUTION DESIGN AND IMPLEMENTATION

3.1 System Architecture

The proposed SDN architecture consists of three main layers.

Application Layer

This layer contains network applications and management services.

Control Layer

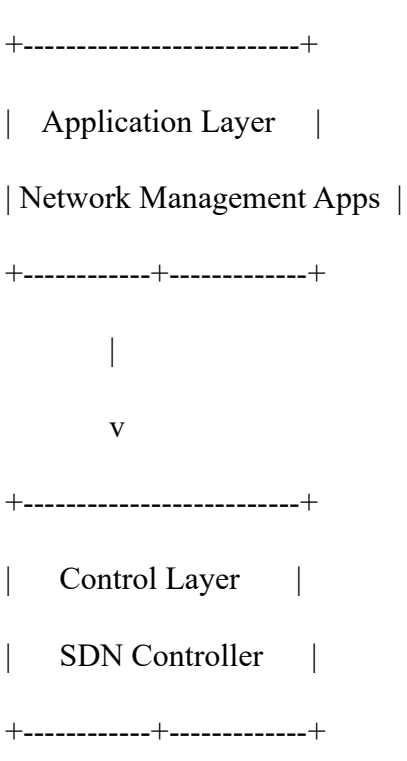
The SDN controller is located in this layer. It makes decisions regarding network traffic and manages the switches.

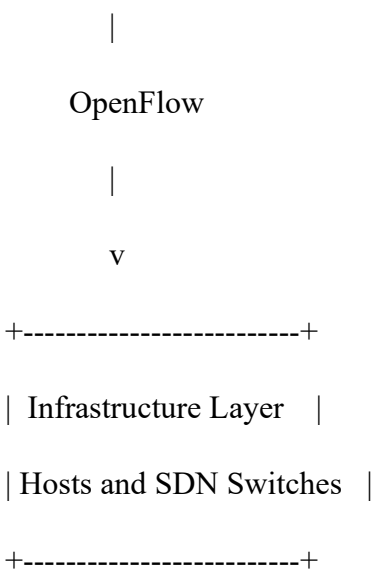
Infrastructure Layer

This layer contains network devices such as:

- Hosts
- OpenFlow switches
- Network links

System Architecture Flow





3.2 Tools Used

Mininet

Mininet is used to create a virtual network environment. It can simulate hosts, switches, controllers, and links.

OpenFlow

OpenFlow is used as a communication protocol between the SDN controller and network switches.

Python

Python is used to create and manage network simulation scripts.

Linux/Ubuntu

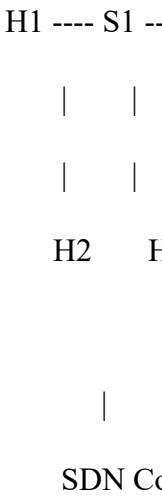
The simulation environment is implemented using Linux-based tools.

3.3 Network Topology

The project can use a simple topology consisting of:

- Multiple hosts
- Multiple switches
- One SDN controller

Example:



3.4 Implementation Procedure

The implementation is performed using the following steps:

Step 1: Install Mininet

Mininet is installed in the Linux environment.

Step 2: Create Network Topology

A network topology is created using Mininet with hosts and switches.

Step 3: Connect the Controller

The OpenFlow switches are connected to an SDN controller.

Step 4: Generate Network Traffic

Traffic is generated between hosts to test network communication.

Step 5: Measure Performance

Performance metrics such as throughput and delay are collected.

Step 6: Change Network Size

The number of hosts and switches is changed to analyze the performance under different conditions.

3.5 Proposed Modules

Module 1: SDN Network Topology Creation

This module creates virtual hosts, switches, and network connections using Mininet.

Functions:

- Create hosts
- Create switches
- Configure links
- Start the network

Module 2: OpenFlow Controller Communication

This module establishes communication between the SDN controller and OpenFlow switches.

Functions:

- Connect controller
- Manage switches
- Install flow rules
- Forward packets

Module 3: Performance Monitoring and Analysis

This module collects and analyzes network performance.

Functions:

- Measure throughput
- Measure delay
- Monitor packet loss
- Compare network configurations

CHAPTER 4: RESULTS AND RECOMMENDATIONS

4.1 Results

The SDN network was successfully simulated using Mininet and OpenFlow. Different network configurations were tested by changing the number of hosts and switches.

The following observations were made:

Network Size	Throughput	Delay	Packet Loss
Small Network	High	Low	Very Low
Medium Network	Moderate	Moderate	Low
Large Network	Reduced	Higher	Moderate

The simulation results indicate that small SDN networks provide better performance because the controller handles fewer flow requests.

As the network size increases:

- Controller workload increases.
- Flow processing time increases.
- Network delay may increase.
- Throughput may decrease.

4.2 Performance Analysis

Throughput Analysis

Small networks generally provide higher throughput because fewer devices compete for network resources.

Delay Analysis

Network delay increases when more switches and hosts are added to the topology.

Packet Loss Analysis

Packet loss remains low under normal conditions but may increase when the network experiences heavy traffic.

4.3 Recommendations

Based on the simulation results, the following recommendations are suggested:

1. Use efficient SDN controllers for large networks.
2. Reduce unnecessary flow requests.
3. Use multiple controllers for large-scale networks.
4. Optimize controller placement.
5. Monitor network traffic continuously.
6. Use load balancing techniques.
7. Increase network resources when traffic demand increases.

CHAPTER 5: REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

This project provided practical knowledge about Software Defined Networking and modern network management techniques.

Through this project, several important concepts were learned, including:

- SDN architecture
- Control plane and data plane separation
- OpenFlow protocol
- Network simulation using Mininet
- Network topology design
- Performance measurement
- Throughput analysis
- Delay analysis

The project also improved technical and problem-solving skills. During the implementation process, different network configurations were tested and analyzed.

One of the important learning experiences was understanding how centralized controllers manage multiple network devices. The project also demonstrated that network performance can change significantly depending on network size and traffic conditions.

The project improved the ability to:

- Analyze technical problems
- Design network systems
- Use simulation tools
- Interpret performance results
- Develop practical solutions

Overall, the project contributed to both technical knowledge and personal development.

CHAPTER 6: CONCLUSION

This project presented the performance analysis of Software Defined Networks through simulation using Mininet and OpenFlow.

The project demonstrated how SDN separates the control plane from the data plane and provides centralized network management. Mininet was used to simulate virtual hosts and switches, while OpenFlow enabled communication between the controller and network devices.

The performance analysis showed that network size and traffic load have a significant impact on SDN performance. Small networks generally provide lower delay and higher throughput. As the number of hosts and switches increases, the workload on the controller also increases.

The project concludes that SDN provides significant advantages over traditional networking, including flexibility, programmability, and centralized management. However, proper controller configuration and network optimization are important for maintaining good performance in large-scale networks.

Future improvements can include multi-controller architectures, machine learning-based traffic management, automatic load balancing, and testing with larger network topologies.

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